

CHE 313 HW #1

100/100

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18.1-1.) A gas of CH₄ and He is contained in a tube at 101.32 kPa pressure and 298 K. At one point, the partial pressure of methane is $P_{A1} = 60.79$ kPa, and at a point .02 m distance away, $P_{A2} = 20.26$ kPa. If the total pressure is constant throughout the tube, calculate the flux of CH₄ (methane) at steady state for equimolar counterdiffusion.

$$J_{Az}^* = -D_{AB} \frac{dc_A}{dz}$$

$$J_{Az}^* dz = -D_{AB} dc_A$$

$$J_{Az}^* \int_{z_1}^{z_2} dz = -D_{AB} \int_{C_{A1}}^{C_{A2}} dc_A$$

$$J_{Az}^* (z_2 - z_1) = -D_{AB} (C_{A2} - C_{A1})$$

$$J_{Az}^* = \frac{D_{AB} (C_{A1} - C_{A2})}{(z_2 - z_1)}$$

$$\Rightarrow J_{Az}^* = \frac{D_{AB} (P_{A1} - P_{A2})}{RT(z_2 - z_1)}$$

$$\begin{aligned} J_{Az}^* &= \frac{(.675 \times 10^{-4} \text{ m}^2/\text{s}) (60.790 \text{ kPa} - 20.260 \text{ kPa})}{\left(8314.3 \frac{\text{N} \cdot \text{m}}{\text{kg} \cdot \text{mol} \cdot \text{K}}\right) (298 \text{ K}) (.02 \text{ m} - 0 \text{ m})} \\ &\approx 5.52 \times 10^{-5} \text{ kg mol A/m}^2 \cdot \text{s} \end{aligned}$$

$$PV = nRT$$

$$\frac{n}{V} = \frac{P}{RT} = C$$

$$60.79 \text{ kPa} \times \frac{1000 \text{ Pa}}{1 \text{ kPa}} = 60,790 \text{ Pa}$$

$$20.26 \text{ kPa} \times \frac{1000 \text{ Pa}}{1 \text{ kPa}} = 20,260 \text{ Pa}$$

$$\text{From Table 18.2-1: } D_{AB} = .675 \times 10^{-4} \text{ m}^2/\text{s}$$

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18.2-1. For a mixture of ethanol vapor and methane, predict the diffusivity using the method of Fuller et al. (F1)

a) At 1.0132×10^5 Pa; 298 K
(1 atm)

$$D_{AB} = \frac{1.00 \times 10^{-7} T (1/M_A + 1/M_B)^{1/2}}{P [(\sum V_A)^{1/3} + (\sum V_B)^{1/3}]^2}$$

$$\sum V_A = 2(16.5) + 6(1.98) + 1(5.48) = 50.36$$

$$M_{\text{ethanol}} = M_A = 46.07 \text{ g/mol}$$

$$\sum V_B = 1(16.5) + 4(1.98) = 24.42$$

$$M_{\text{methane}} = M_B = 16.04 \text{ g/mol}$$

$$D_{AB} = \frac{1.00 \times 10^{-7} (298)^{1.75} (1/46.07 + 1/16.04)^{1/2}}{1.0 [(50.36)^{1/3} + (24.42)^{1/3}]^2} = 1.43 \times 10^{-5} \text{ m}^2/\text{s} // \checkmark$$

$$D_{AB} = \frac{1.00 \times 10^{-7} (373)^{1.75} (1/46.07 + 1/16.04)^{1/2}}{1.0 [(50.36)^{1/3} + (24.42)^{1/3}]^2} = 2.11 \times 10^{-5} \text{ m}^2/\text{s} // \checkmark$$

b) At 2.0265×10^5 Pa; 298 K

$$D_{AB} = \frac{1.00 \times 10^{-7} (298)^{1.75} (1/46.07 + 1/16.04)^{1/2}}{2 [(50.36)^{1/3} + (24.42)^{1/3}]^2} = 7.13 \times 10^{-6} \text{ m}^2/\text{s} // \checkmark$$

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(8.2-5.) Dilute NaOH is diffusing in an aqueous solution. Do as follows:

a) Estimate the diffusivity of NaOH at 25°C. Also, calculate the diffusivity of the individual ions Na^+ & OH^- .

$$D_{AB}^0 = 8.928 \times 10^{-10} (298) \frac{(1/1 + 1/1)}{(1/50.1 + 1/197.6)} \\ \approx 2.127 \times 10^{-5} \text{ cm}^2/\text{s} //$$

$$\text{Eqn } D_{AB}^0 = 8.928 \times 10^{-10} T \frac{(1/n_+ + 1/n_-)}{(1/\lambda_+ + 1/\lambda_-)}$$

From Table 18.2-5:

$$\text{OH}^-: \lambda^0 = 197.6$$

$$\text{Na}^+: \lambda^0 = 50.1$$

$$\text{Eqn (8.2-16)} \quad D_i = 2.662 \times 10^{-7} \frac{\lambda_i}{n_i}$$

$$D_{\text{Na}^+} = 2.662 \times 10^{-7} \frac{50.1}{1} = 1.33 \times 10^{-5} \text{ cm}^2/\text{s} \quad \checkmark$$

$$D_{\text{OH}^-} = 2.662 \times 10^{-7} \frac{197.6}{1} = 5.26 \times 10^{-5} \text{ cm}^2/\text{s} \quad \checkmark$$

b) Estimate the diffusivity of NaOH at 15°C

$$T/334 \mu_{\text{NaOH}}$$

$$(288.15) / (334 \times 1.141) \\ \approx 0.756 \text{ K/cp}$$

$$D_{AB}^{15^\circ\text{C}} = D_{AB}^{25^\circ\text{C}} \times 0.756 = 2.127 \times 10^{-5} \times 0.756 \\ \text{cm}^2/\text{s}$$

$$\approx 1.608 \times 10^{-5} \text{ cm}^2/\text{s} //$$

Interpolation for 15°C (values from Table A.2-4)

$$\mu_{15^\circ\text{C}} = \frac{(T_{15^\circ\text{C}} - T_{14^\circ\text{C}}) (\mu_{16^\circ\text{C}} - \mu_{14^\circ\text{C}})}{(T_{16^\circ\text{C}} - T_{14^\circ\text{C}})} + \mu_{14^\circ\text{C}}$$

$$= \frac{(288.15 - 287.15) (1.1111 - 1.1709) \text{ cp}}{(289.15 - 287.15 \text{ K})} + 1.1709 \text{ cp}$$

$$= 1.14 \text{ cp}$$

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18.4-1.) Prove the following equation using definitions in Table 18.4-1: $N_A + N_B = C v_m$

$$N_A = J_A^* + C_A v_m$$

$$J_A^* = C_A (v_A - v_m)$$

$$J_A^* = -J_B^* \checkmark$$

$$\Rightarrow J_B^* = -J_A^* \checkmark$$

$$N_B = J_B^* + C_B v_m$$

$$J_B^* = C_B (v_B - v_m)$$

$$N_A = C_A v_A$$

$$N_B = C_B v_B$$

Proof

$$N_A + N_B = C v_m$$

$$J_A^* + C_A v_m + J_B^* + C_B v_m = C v_m$$

$$J_A^* + C_A v_m + (-J_A^*) + C_B v_m = C v_m$$

$$C_A v_m + C_B v_m = C v_m \checkmark$$

$$\Rightarrow N_A + N_B = C v_m \checkmark$$

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18.4-5.) Gas A is diffusing from a gas stream at point 1 to a catalyst surface at point 2, and reacts instantaneously and irreversibly as follows: $2A \rightarrow B$

Gas B diffuses back to the gas stream. Derive the final equation for N_A at constant pressure P and steady state in terms of partial pressures.

$$20/30 \quad N_A + N_B = C v_m$$

$$\Rightarrow v_m = \frac{N_A + N_B}{C}$$

$$N_A = J_A^* + C_A v_m$$

$$X_A = C_A / C$$

$$\Rightarrow N_A = J_A^* + \frac{C_A}{C} (N_A + N_B)$$

$$J_A^* = -C D_{AB} \frac{dx_A}{dz}$$

$$\Rightarrow N_A = -C D_{AB} \frac{dx_A}{dz} + \frac{C_A}{C} (N_A + N_B)$$

$$\Rightarrow N_A = -C D_{AB} \frac{dx_A}{dz} + X_A (N_A + N_B)$$

$$N_A = -C D_{AB} \frac{dx_A}{dz} + X_A (N_A + N_B)$$

$$N_A - X_A (N_A + N_B) = -C D_{AB} \frac{dx_A}{dz}$$

$$\Rightarrow \frac{-dx_A}{N_A - X_A (N_A + N_B)} = \frac{dz}{C D_{AB}}$$

1)

$$\int_{x_{A1}}^{x_{A2}} \frac{1}{N_A - x_A(N_A + N_B)} dx_A = \frac{1}{CD_{AB}} \int_{z_1}^{z_2} dz$$

$$\int_{x_{A1}}^{x_{A2}} \frac{1}{N_A - x_A(N_A + N_B)} dx_A = \frac{(z_2 - z_1)}{CD_{AB}}$$

Let $u = N_A - x_A(N_A + N_B)$

$$du = -(N_A + N_B) dx_A$$

$$-\frac{1}{(N_A + N_B)} \int_{x_{A1}}^{x_{A2}} \frac{1}{N_A - x_A(N_A + N_B)} dx_A = \frac{(z_2 - z_1)}{CD_{AB}}$$

2A → B
const pressure, steady state, B diffuses back to state

$$N_A = 2N_B$$

$$\Rightarrow N_B = \frac{-N_A}{2}$$

$$\frac{1}{(N_A + N_B)} \int \frac{1}{u} du = \frac{(z_2 - z_1)}{CD_{AB}}$$

$$\frac{N_A}{N_A + N_B} = \frac{N_A}{N_A - \frac{N_A}{2}} = \frac{N_A}{\frac{N_A}{2}} = 2$$

$$\frac{N_A}{2} = \frac{2N_A}{N_A} = 2$$

$$\frac{1}{(N_A + N_B)} \ln |u| \Big|_{x_{A1}}^{x_{A2}} = \frac{(z_2 - z_1)}{CD_{AB}}$$

$$\frac{1}{(N_A + N_B)} \ln |N_A - x_A(N_A + N_B)| \Big|_{x_{A1}}^{x_{A2}} = \frac{(z_2 - z_1)}{CD_{AB}}$$

@ const P, assume perfect gas law
 $PV = nRT$

$$\Rightarrow \frac{n}{V} = \frac{P}{RT} = C$$

$$\left(\frac{1}{N_A} \right) \left(\frac{N_A}{N_A + N_B} \right) \ln \left| \frac{N_A}{N_A + N_B} - x_A \right| \Big|_{x_{A1}}^{x_{A2}} = \frac{(z_2 - z_1)}{CD_{AB}}$$

$$\Rightarrow \left(\frac{1}{N_A} \right) \left(\frac{N_A}{N_A + N_B} \right) \ln \left| \frac{N_A}{N_A + N_B} - x_A \right| \Big|_{x_{A1}}^{x_{A2}} = \frac{(z_2 - z_1)}{CD_{AB}}$$

$$N_A = \frac{2D_{AB}P}{RT(z_2 - z_1)} \ln \left| \frac{1 - \frac{P_{A2}}{2P}}{1 - \frac{P_{A1}}{2P}} \right|$$

$$\Rightarrow \left(\frac{1}{N_A} \right) 2 \ln \left| \frac{2 - x_{A2}}{2 - x_{A1}} \right| = \frac{z_2 - z_1}{CD_{AB}}$$

$$\Rightarrow N_A = \frac{2CD_{AB}}{z_2 - z_1} \ln \left| \frac{2 - \frac{C_{A2}}{C}}{2 - \frac{C_{A1}}{C}} \right| \Rightarrow N_A = \frac{2D_{AB}P}{RT(z_2 - z_1)} \ln \left| \frac{2 - \frac{P_{A2}}{P}}{2 - \frac{P_{A1}}{P}} \right|$$